

National University of Computer & Emerging Sciences

Course Outline: EE-2003 Computer Organization and Assembly Language

Program: BS Computer Science/Software Engineering/Artificial Intelligence

Semester: Fall 2026

Credit Hours: 3+1

Pre-requisite(s): EE1005 Digital Logic Design

Instructor: Dr. Usman Abbasi/ Dr. Maqsood M. Khan/Engr. Usman Malik

Course Description:

The main objective of this course is to introduce the organization of computer systems and usage of assembly language for optimization and control. Emphasis should be given to expose the low-level logic employed for problem solving while using assembly language as a tool. This course offers a practical, applied introduction to computer organization and 16-bit x86 assembly using Intel 8086/8088 architecture. Students will explore how instructions interact with registers, segment registers, memory addressing modes, the stack and calling conventions, control flow, bitwise and shift operations, string instructions, I/O port programming (IN/OUT), and BIOS/DOS interrupts. Emphasis is placed on writing and debugging assembly programs, integrating assembly subroutines, and understanding performance and optimization at the machine level. All programming and labs will be conducted in a consistent and portable environment using DOSBox with a classic x86/88 assembler and tools.

By the end of the course, students will not only understand how computers execute at the machine level but will also be able to apply these skills in practical computing and problem-solving scenarios. Students will learn to write .ASM code, assemble and link it with TASM/MASM, run and debug programs in DOSBox, trace registers and flags, and optionally integrate assembly routines with higher-level languages such as C.

Text & Reference Book(s):

Text: Assembly Language for Intel-Based Computers (7th Ed) by Kip R. Irvine published by Pearson Education Inc. (ISBN 978-0-07-338065-0)

Computer System Architecture, M. Morris Mano, Latest Edition

Reference: Assembly Language Programming and Org. of the IB by Ytha Yu, Charles Marut Published by McGraw Hill

Reference Material: Assembly Language Programming lectures notes delivered by Belal Hashmi

Assessment Plan:

Quizzes (6)	12 %
Assignments (4)	08%
Sessional I	15 %
Sessional II	15 %
Final exam	50%
Total	100%

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Program Learning Outcomes:

Students are expected to achieve the following PLOs in this course.

PLO 2: Knowledge for Solving Computing Problems: Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.

PLO 4: Design/ Development of Solutions: Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

Sustainable Development Goals (SDGs):

SDG 4 Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 8 Decent Work and Economic Growth: Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

SDG 9 Industry, Innovation, and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

Course Learning Outcomes:

Upon successful completion of this course, students will be able to:

CLO	Name	Domain	Taxonomy Level	PLO	Assessment Tools
1	Demonstrate the basic concepts of computer organization including CPU, memories, and input/output and explain their purposes and interactions.	Cognitive	C3	PLO-2	A1, S1, F1, Q1, Q2, F2
2	Describe the working of important x86/88 assembly primitives, including arithmetic, branching, bit manipulation, addressing modes and interrupt handling.	Cognitive	C2	PLO-2	A2, A3, Q3, Q4, S1, S2, F3, F4
3	Apply the knowledge of Intel x86/88 architecture to develop moderately complex and well-modularized assembly programs.	Cognitive	C4	PLO-4	A4, Q5, Q6, F5, F6

A = Assignment, Q = Quiz, S = Sessional, F = Final, P = Project, W = Written Report

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Course Contents:

List of Topics	Week	CLOs
Course Introduction: Introduction to Assembly Language and Computer Organization, Computer Architecture vs Computer Organization, Introduction to Basic Concepts, Basic Computer Architecture, Memories Organization and capacity	1	1
Intel iapx86 Architecture: Registers, Instruction Groups, History, Register Architecture, Our First Program, Segmented Memory Model, Assembly Language Fundamentals: Basic Language Elements, Assembling Linking and Running Programs (Book1-CH3)	2	1
Addressing Modes: Data Declaration, Direct Addressing, Size Mismatch Errors, Register Indirect Addressing, Register + Offset Addressing, Segment Association, Address Wraparound	3	2
Branching: Comparison and Conditions, Conditional Jumps, Unconditional Jump,	4	2
Branching (Continue): Relative Addressing, Types of Jump, Sorting Example)	5	2
<i>WEEK 6</i>		
Bit Manipulations: Multiplication Algorithm, Shifting and Rotations, Multiplication in Assembly Language.	7	3
Bit Manipulations (Continue): Extended Operations, Bitwise Logical Operations, Masking Operations	8	3
Subroutines: Program Flow, Our First Subroutine, Stack, Saving and Restoring Registers,	9	3
Subroutines (Continue): Parameter Passing Through Stack, Local Variables.	10	3
<i>Week 11</i>		
Display Memory: ASCII Codes, Display Memory Formation, Hello World in Assembly Language, Number Printing in Assembly, Screen Location Calculation.	12	1,3
String Instructions: String Processing, STOS Example – Clearing the Screen, LODS Example – String Printing, SCAS Example – String Length, LES and LDS Example, MOVS Example – Screen Scrolling, CMPS Example – String Comparison.	13	2,3
Software Interrupts: Interrupts, Hooking an Interrupt, BIOS and DOS Interrupts.	14	3
Real Time Interrupts and Hardware Interfacing: Hardware Interrupts, I/O Ports, Terminate and Stay Resident, Programmable Interval Timer, Parallel Port. Debug Interrupts: Debugger using single step interrupt, Debugger using breakpoint interrupt.	15	3
Revisions	16	